

AXIOM AI (Tutor AI) - Minimum Viable Product (MVP) Definition (Revised)

Project: AXIOM AI (Multi-Agent SaaS for Sri Lankan Shadow Education)
Document Type: MVP Scope & Feature Definition
Objective: To define a strict, streamlined Minimum Viable Product that solves the core bottlenecks (communication chaos and resource distribution) within a 3-to-4 week development cycle, utilizing a simplified architecture and focusing on essential manual handoffs before scaling to full automation.

1. The Core Problem vs. The MVP Solution

Sri Lankan private tutors manage hundreds to thousands of students via unstructured messaging groups. The primary pain points are repetitive questions, manual onboarding, and tracking student context.
The MVP Solution: A Twilio-integrated digital agent that automatically profiles new students, instantly retrieves educational resources via a Google Drive connection, and intelligently hands off payment verification and complex queries to a human tutor via a dedicated staff dashboard.

2. Scope Definition (What's In vs. What's Out)

To ensure rapid deployment and minimize architectural overhead, we are pivoting to Twilio and keeping memory and dashboards lightweight.

Included in MVP (The "Must Haves")	Excluded from MVP (v2.0 & Beyond)
Twilio Messaging Integration (Ingestion & Replies)	Meta WhatsApp Business API & Telegram Bot API
Short-Term (ST) & Procedural Memory	Long-Term (LT) Semantic & Episodic Memory
Payment Escalation Queue (Manual Staff	Vision Model (OCR) for Automated Bank

Included in MVP (The "Must Haves")	Excluded from MVP (v2.0 & Beyond)
Verification)	Slip Reconciliation
MVP Staff Dashboard (Classes, Logs, Overview, Chat)	Complex Web CRM features and Automated MCQ Grading
Admissions & Resource RAG Agents	Voice Note Transcription & Processing

3. MVP Core Feature Deep Dive

Feature 1: Twilio Messaging Gateway (Reduced Overhead)

- **The Problem:** Strict webhook timeout constraints (like Meta's 3-second rule) inflate backend architecture, requiring complex message queues and retry handling.
- **The MVP Solution:** Utilizing Twilio allows for a simpler API layer. FastAPI receives the message and processes the LangGraph workflow with reduced infrastructure overhead, maintaining stable communication.

Feature 2: The Admissions & Onboarding Agent

- **The Problem:** Tutors waste hours manually adding student details.
- **The MVP Solution:** Using procedural memory rules ("When a student is admitted, do X"), the Admissions Agent extracts the student's Name, Grade, and Subject, saving them to the database and sending introductory materials.

Feature 3: The Resource Retrieval Agent (Drive RAG)

- **The MVP Solution:** An agent equipped with an MCP server connected to Google Drive retrieves specific past papers or lecture links instantly upon student request.

Feature 4: Payment Check Agent & Staff Dashboard

- **The Problem:** Automated vision OCR is too risky/complex for MVP, but payments and escalations must be handled smoothly.
- **The MVP Solution:** When a student uploads a payment receipt, the Payment Check

Agent routes it directly to the Staff Dashboard for manual review. The MVP Dashboard enables staff to:

- Monitor classes and add new ones.
- View a high-level data overview and full message logs.
- Use a separate chat interface to manually take over conversations, confirm payments, or resolve frustrated escalations.

4. MVP Technical Architecture & Tooling

- **API & Webhooks:** FastAPI (Python) connected to Twilio.
- **AI Orchestration:** LangGraph (StateGraph for routing).
- **Memory System:** Restricted to **Short-Term (ST) Memory** (session context) and **Procedural Memory** (workflow rules).
- **Database:** Supabase (PostgreSQL) for Relational Student Data, Classes, and Chat Logs.
- **Frontend:** React-based MVP Staff Dashboard for monitoring, class management, and live chat intervention.

5. Success Metrics for the MVP

1. **Deflection Rate:** The AI successfully handles >75% of incoming routine messages.
2. **Escalation Efficiency:** Staff can confirm payments or take over a chat within the dashboard in under 3 clicks.
3. **System Stability:** Seamless message delivery via Twilio with no webhook timeouts or dropped context.

6. Team Feedback Matrix (Action Required)

Instructions: Each of the 5 team members should claim a row below. Document any architectural red flags, scope creep concerns, database schema tweaks, or general suggestions regarding this MVP scope before we kick off development.

Team Member	MVP Component / Subject	Feedback / Fixes / Suggestions	Status
[Name 1]	e.g., Twilio Integration	[Enter your notes here...]	Open

Team Member	MVP Component / Subject	Feedback / Fixes / Suggestions	Status
[Name 2]	e.g., MVP Dashboard UI	[Enter your notes here...]	Open
[Name 3]	e.g., Memory Constraints	[Enter your notes here...]	Open
[Name 4]	e.g., Payment Escalation Flow	[Enter your notes here...]	Open
[Name 5]	e.g., General Scope	[Enter your notes here...]	Open